

Indian Institute of Technology, Gandhinagar
ES226 Semiconductor Devices
Semester 2, 2025-2026
Tutorial - 4

Q1. Read Section 3.3 (Carrier Concentrations) from Streetman book, Read Section 2.5 (Intrinsic Semiconductor), 2.6 (Extrinsic Semiconductor) from “Fundamentals of Semiconductor Devices” by B. Anderson & R. Anderson – Try to understand the donor level wrt E_c

Finding Carrier Concentration:

- a. What is the density of states (DOS) i.e. $g(E)$? What is the relation of DOS with Energy? Plot E vs $g(E)$.
- b. Find an expression for the total number of available states per cm^3 in the conduction band between energies E_c and $E_c + aKT$. Where a is the arbitrary constant.
- c. Why does an electron have an occupancy probability ($f(E)$)?
- d. Plot E vs $f(E)$ and also plot $f(E)$ vs E for visualization in different ways.
- e. What is the Fermi level?
- f. How does probability change with temperature? Describe the probability below and above Fermi level energy for $T_1 = 0K$, $T_2 > T_1$, and $T_3 > T_2$
 - i. How to find carrier concentration $n(E)$ or $p(E)$ using $g(E)$ and $f(E)$? How to find total carrier concentration n or p ?
- g. Graphically show the band diagram, DOS, occupancy probability, and carrier concentration for an intrinsic and extrinsic semiconductor.
- h. Find carrier concentration from DOS and $f(E)$ for an intrinsic semiconductor in the valence band and conduction. What is the electron population in a non-degenerate semiconductor at $E = E_c + 5kT$, You can assume the material to be Silicon?

Q2. A hypothetical semiconductor has an intrinsic Carrier concentration of 10^{10} per cm^3 at 300K and it has conduction band and valence band effective densities of states N_c and N_v , both equal to 10^{19} per cm^3 .

- a. What is the Energy band gap (E_g)
- b. If the semiconductor is doped with $N_d = 10^{16} \text{ cm}^{-3}$ and $N_a = 2 \times 10^{16} \text{ cm}^{-3}$. Determine Electron and Hole concentration under equilibrium at 300 K
- c. For part (b), Draw an Energy band diagram and compute $E_f - E_i$ in terms of eV

Q3. Temperature Dependence of n_i

- a) Plot n_i^2 vs $1/T$ for Si. Derive that slope $= -E_g/k$.
- b) An n-type Si resistor ($N_d = 10^{15} \text{ cm}^{-3}$) is used as a temperature sensor. At what temperature will n_i exceed N_d by 10%? What happens to resistivity at this point?

c) Why is wide-bandgap semiconductor (e.g., GaN, $E_g = 3.4$ eV) preferred for high-temperature power electronics compared to Si?

Q4. For a Silicon material, Find

- (a) Effective number of atoms ?
- (b) Atomic and Volume density for (001) plane?

Q5. Read Section 3.2 Drift Current and 3.3 Carrier Mobility From “Fundamentals of Semiconductor Devices – Anderson book”

- a. Derive the expression for drift current in a n-type semiconductor – is the current given only by electrons?
- b. Is drift current the physics behind Ohm’s law? Under what condition?
- c. What is conductivity and resistivity?
- d. How to find resistance?
- e. Can we find electron and hole resistance separately? Is the total current a series combination of electron and hole resistance or parallel combination?
- f. Define separate electron and hole movement direction for an applied Voltage to a n-type semiconductor. Also, define direction of electron, hole current and total current?
- g. Mobility:
 - a. What is mobility? Is hole mobility higher or electron mobility?
 - b. Describe different types of scattering that affects mobility? Also describe their temperature dependence.
 - c. Describe effective mobility and plot effective mobility with temperature (See book Fig. 3.8)
- h. Solve the room temperature resistivity of intrinsic silicon and compare both that with the uncompensated n- and p-type silicon doped with 10^{17} cm⁻³ donors (or acceptors). Assume $n=n_0$ and $p=p_0$.
- i. What is high field saturation?
 - a. Is Ohm’s law valid in this case?
 - b. Plot velocity vs E-field. How does this plot change if mobility changes?

Q6. A sample of Germanium is made p-type by adding Indium at the rate of one indium atom for every 2.5×10^8 germanium atoms. Given $n_i = 2.5 \times 10^{19}$ /m³ at 300 K and the number of germanium atoms per m³ = 4.4×10^{28} . What is the value of n_p ?

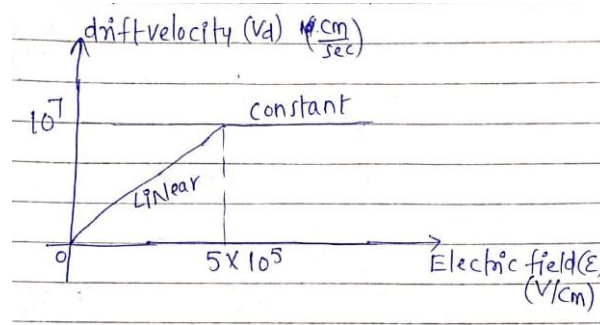
Q7. A silicon sample is non-uniformly doped n-type with $N_d = 10^{15}$ per cm³. Draw an Energy band diagram under equilibrium condition.

Q8. The current measured across a Si bar, when 10V is applied at 300K, is found to be 1.12mA. It is given that cross sectional area of this bar is 100 μm^2 . The phosphorus doping is 10^{17} cm⁻³. Determine the length of the Si bar. Take μ_n as 700 cm²/V-s.

Q9. A 2 cm long piece of silicon with cross-sectional area of 0.1 cm² is doped with donors at 10^{15} cm⁻³ is maintained at T=300K and has a resistance of 90 Ω . The saturation velocity of electrons in the silicon is 10^7 cm/sec for fields above 10^5 V/cm. Calculate the electron drift velocity if we apply

a voltage of 100 V across the piece. What is the current through the piece if the applied voltage of 10^6 V across it.

Q10. The dependence of drift velocity of electrons on an electric field in a semiconductor is shown below. The semiconductor has a uniform electron concentration of 10^{16} cm^{-3} and the charge of electron $q = 1.6 \times 10^{-19} \text{ C}$. If a bias of 5 volts is applied across a $1 \text{ } \mu\text{m}$ region and cross-sectional area $0.01 \text{ } \mu\text{m}^2$ of this semiconductor. Determine the current flowing through the semiconductor?



Q11. The electron concentration in a sample of uniformly doped n-type silicon at 300K varies linearly from 10^{17} cm^{-3} at $x = 0$ to $6 \times 10^{16} \text{ cm}^{-3}$ at $x = 2 \text{ } \mu\text{m}$. If the charge of electrons is $1.6 \times 10^{-19} \text{ C}$ and the diffusion coefficient of electrons is $35 \text{ cm}^2/\text{sec}$. Find the current density of silicon, if no electric field is present in the silicon.

Q12. The electron concentration in silicon at $T = 300 \text{ K}$ is given by

$$n(x) = 10^{16} \exp\left(\frac{-x}{18}\right) \text{ cm}^{-3}$$

where x is measured in μm and is limited to $0 < x < 25 \text{ } \mu\text{m}$. The electron diffusion coefficient is $D_n = 25 \text{ cm}^2/\text{s}$ and the electron mobility is $\mu_n = 960 \text{ cm}^2/\text{V-s}$. The total electron current density through the semiconductor is constant and equal to $J_n = 40 \text{ A/cm}^2$. The electron current has both diffusion and drift current components. Determine the electric field as a function of x which must exist in the semiconductor.

Q13. Electrons in undoped gallium arsenide have a mobility of $8,800 \text{ cm}^2/\text{V-s}$. Calculate the average time between collisions. Calculate the distance travelled between two collisions (also called the mean free path). Use an average velocity of 10^7 cm/s .

Q14. Calculate the electron and hole densities in an n-type silicon wafer ($N_d = 10^{17} \text{ cm}^{-3}$) illuminated uniformly with 10 mW/cm^2 of red light ($E_{ph} = 1.8 \text{ eV}$). The absorption coefficient of red light in silicon is 10^{-3} cm^{-1} . The minority carrier lifetime is 10 ms .

Q15. An engineer has two semiconductor bars of identical dimensions one made of Germanium (Ge) and one of Gallium Arsenide (GaAs). Both are doped with the same concentration of donors, $N_d = 10^{16} \text{ cm}^{-3}$.

- Which bar will have a higher concentration of minority carriers (holes) at 300K?
- At what approximate temperature would the Ge bar start to behave intrinsically (when n_i becomes comparable to N_d)? Will this temperature be higher or lower for the GaAs bar? Justify your answer based on the bandgaps of the materials.

(Use: For Ge, $E_g = 0.66 \text{ eV}$, $n_i = 2.4 \times 10^{13} \text{ cm}^{-3}$. For GaAs, $E_g = 1.42 \text{ eV}$, $n_i = 2.1 \times 10^6 \text{ cm}^{-3}$).

Q16. Near the bottom of the conduction band in a semiconductor, the electron energy can be expressed as $E = E_c + AK^2$, where A is independent of K and is positive.

- Find an expression for the electron effective mass in terms of A .
- Find an expression for the electron velocity as a function of E .

Q17. Read Section 3.2 (Drift Current), 3.3. (Mobility) – for detailed understanding of mobility, 3.4 (Diffusion Current) from “Fundamentals of Semiconductor Devices” by B. Anderson & R. Anderson

You may also refer 3.4 (Drift of carriers in Electric and Magnetic Field) for more details, 4.4. (Diffusion of carriers) from Streetman book

Derive an expression for drift and diffusion current

Q18. Read Section 4.2 (Graded Doping) from “Fundamentals of Semiconductor Devices” by B. Anderson & R. Anderson

Derive the relationship between diffusion coefficient and mobility. What is it known as, and what does it signify?

Q19. Write the units of various parameters listed below.

Density of states, Probability Function, carrier concentration, Resistivity, Conductivity, Mobility, Resistance, Flux, Velocity, Momentum, wavelength, bandgap, mass, effective mass, lattice constant, Charge, volume charge density, surface charge density, Current, Current density, Voltage, Band-diagram energies (E_c , E_v , E_i , E_f), Electric Field, electron affinity, permittivity.

Q20. For a Silicon atom:

Read Section 1.3.3 (Quantum Number and Pauli's exclusion principle), and 1.3.4 (Covalent Bonding in Crystalline Solid) from the book “Fundamentals of Semiconductor Devices” by B. Anderson & R. Anderson

- How many electrons does Silicon have?
- Do all of the electrons participate in co-valent bond? Which electrons participate?
- Draw the potential energy for the simple 2D silicon lattice with co-valent bonds and show all the electrons positions? Which electrons will be bound to their own nucleus and which electrons are shared with another nucleus?
- How many neighbours does each atom have? Will electrons get influenced by nearest nucleus only?
- Imagine how the potential energy would look in 1D, 2D
- How many states will be present in this arrangement. Draw potential energy diagram.
- Can we see all these states separately? How are the bands formed? Will there be any band above conduction band? Is there any band below valence band?
- Do these bands and the gap between them change if we change the material? Why?

Diffusion Current is part of the tutorial 4 and will be covered in Tuesday's class.

You may refer to these online lectures to get started early if other questions are answered.

1. https://www.youtube.com/watch?v=Xme87NxXN_8&list=PLgMDNELGJ1CaNcuuQv9xN07ZWkXE-wCGP&index=12
2. <https://www.youtube.com/watch?v=Uh5jBDOFoCk&list=PLgMDNELGJ1CaNcuuQv9xN07ZWkXE-wCGP&index=13>